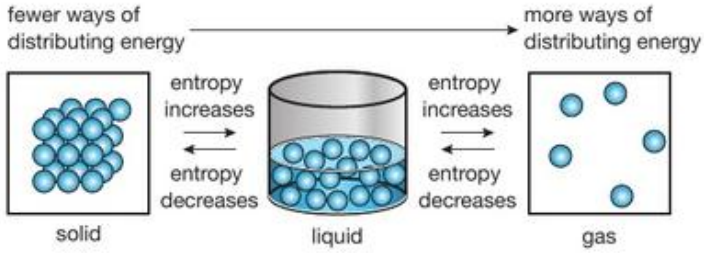
LOOKING AHEAD – ENTROPY

Many reactions we meet at International GSCE level are exothermic, for example combustion and neutralisation. Despite requiring a small amount of energy to get the reactions started (*activation energy*), these reactions release heat energy overall because the products are more stable than the reactants. This shows that enthalpy is certainly a driving force for a reaction to occur. However, endothermic processes do occur. How do they happen? Dissolving ammonium nitrate into water causes the temperature of the solution to decrease, but it dissolves anyway. This is possible because of a different factor called entropy. *Entropy is a measure of the number of ways for the energy in a system to be distributed*. It is greater in a more disordered system because the energy can be dispersed more flexibly. It follows that solids have lower entropy than liquids, which have lower entropy than gases, as shown in Figure 19.18.



▲ Figure 19.18 Entropy is greater in a more disordered system.

If the entropy of a system increases during a reaction, it is more likely for that reaction to occur. When dissolving ammonium nitrate in water, although the process is endothermic, the NH_4^+ and NO_3^- ion particles in the giant ionic lattice become separated from each other. The ions have more freedom of movement in water and the entropy of the system increases, which outweighs the effect of enthalpy.

The reaction between baking soda and vinegar is another example of an endothermic reaction that occurs spontaneously. The reaction produces a gas which leads to an increase in the entropy of the system.



▲ Figure 19.19 The reaction between baking soda and vinegar is used to make papier mâché volcanos.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING, PROBLEM SOLVING (9)

1 a Explain what is meant by an *exothermic reaction* and write balanced chemical equations for any two exothermic changes (apart from the combustion of heptane given in part b).

CHEMISTRY ONLY

SKILLS INTERPRETATION (10)

b Heptane, C_7H_{16}, is a hydrocarbon found in petrol. The equation for the combustion of heptane is:

$$C_7H_{16}(l) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(l) \qquad \Delta H = -4817\,kJ/mol$$

Draw an energy level diagram to show the combustion of heptane. Label the axes and show clearly the reactants, the products and the enthalpy change of reaction on your diagram.

SKILLS REASONING (9)

c Explain in terms of breaking and making bonds why the reaction in b is exothermic.

END OF CHEMISTRY ONLY

(7)

2 a Explain what is meant by an *endothermic reaction*.

CHEMISTRY ONLY

SKILLS INTERPRETATION (10)

b Nitrogen and oxygen gases can combine to form nitrogen monoxide as follows:

$$N_2(g) + O_2(g) \rightarrow 2NO(g) \qquad \Delta H = +180\,kJ/mol$$

Draw an energy level diagram for this process. Label the axes and show clearly the reactants, the products and the enthalpy change of reaction on your diagram.

END OF CHEMISTRY ONLY

SKILLS CRITICAL THINKING (9)

3 Identify each of the following changes as exothermic or endothermic. In some cases you will have to rely on your previous knowledge of chemistry. Several reactions are likely to be completely new to you.

a The reaction between sodium and water.

b Burning methane (major constituent of natural gas).

c The reaction between sodium carbonate and ethanoic acid. A thermometer placed in the reaction mixture shows a temperature drop.

d $S(s) + O_2(g) \rightarrow SO_2(g) \qquad \Delta H = -297\,kJ/mol$

e $CuSO_4(s) \rightarrow CuO(s) + SO_3(g) \qquad \Delta H = +220\,kJ/mol$

f If you dissolve solid sodium hydroxide in water, the solution becomes very hot.

SKILLS INTRAPERSONAL (11)

4 Self-heating cans are used to provide warm food in situations where it is inconvenient to use a more conventional form of heat. By doing an internet (or other) search, find out how self-heating cans work. Write a short explanation of your findings (not exceeding 200 words). You should include equation(s) for any reaction(s) involved, and a diagram or picture if it is useful.

CHEMISTRY ONLY

SKILLS REASONING, PROBLEM SOLVING

10

5 Use the bond energies in the table to calculate the amount of heat released or absorbed when the following reactions take place. In each case, say whether the change is exothermic or endothermic.

Bond	Bond energy/kJ/mol
C–H	413
C–Br	290
Br–Br	193
H–Br	366
H–H	436
Cl–Cl	243
H–Cl	432
O=O	498
O–H	464
N≡N	944
N–H	388

a $CH_4(g) + Br_2(g) \rightarrow CH_3Br(g) + HBr(g)$

(The structure of CH_3Br is the same as that of CH_3Cl)

b $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

c $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$

d $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

END OF CHEMISTRY ONLY

SKILLS EXECUTIVE FUNCTION

4

6 A student investigated the amount of heat given out when hexane, C_6H_{14}, burns using the apparatus in Figure 19.9. Hexane is a highly flammable liquid which is one of the components of petrol (gasoline).

In each case, the student calculated the amount of heat evolved per mole of hexane. Her first two experiments produced answers of –3200 and –3900 kJ/mol of heat evolved. She then decided to do a third experiment. Her results were as follows:

volume of water in copper calorimeter/cm^3	100
mass of burner + hexane before experiment/g	35.62
mass of burner + hexane after experiment/g	35.23
original temperature of water/°C	19.0
final temperature of water/°C	55.0

a Suggest a reason why the student decided to do a third experiment.

5

b Apart from wearing eye protection, suggest two other safety precautions the student should take during the experiment.